

The set of context-free languages, like the set of regular languages, is **closed** under the operations of

- Union
- Concatenation, and
- Kleene *

However, **unlike** the set of regular languages, the set of CFLs is **not closed** under

- Intersection, or
- Difference

Nevertheless, the set of **DCFLs** is still **closed** under Complement. Moreover, the set of CFLs is also **closed** under Intersection with Regular Languages.

Two CFLs Whose Intersection is Not a CFL

$A^nB^nC^n$ can be written as

$$\{a^i b^j c^k \mid i = j \text{ and } j = k\} = \{a^i b^i c^k \mid i, k \geq 0\} \cap \{a^i b^j c^j \mid i, j \geq 0\}$$

A CFL Whose Complement Is Not a CFL

The language

$$XX = \{xx \mid x \in \{a, b\}^*\}$$

is **not** a context-free language, however, its **complement** is a CFL.

Let L be the complement $\{a, b\}^* - XX$. L contains every string in $\{a, b\}^*$ of odd length. If $x \in L$ and $|x| = 2n$ for some $n \geq 1$, then **for some** k with $1 \leq k \leq n$, the k^{th} and $(n + k)^{th}$ symbols of x are different (say a and b , respectively).

There are $k - 1$ symbols before the a , $n - 1$ symbols between the two, and $n - k$ symbols after the b .

- But instead of thinking of the $n - 1$ symbols between the two as $n - k$ and then $k - 1$ (the remaining symbols in the first half, followed by the symbols in the second half that precede the b), we can think of them as $k - 1$ and then $n - k$.

In other words, x is the concatenation of two odd-length strings, the first with a in the middle and $k - 1$ symbols on either side, and the second with b in the middle and $n - k$ symbols on either side. Conversely, every concatenation of two such odd-length strings is in L .

The conclusion is that L can be generated by a CFG.

Therefore, L is a context-free language whose complement is not a CFL.

Please do remember that

- The set of CFLs is *closed* under
 - *Intersection, with Regular Languages*

If L_1 is a context-free language and L_2 is a regular language, then $L_1 \cap L_2$ is a CFL.

- The set of DCFLs is *closed* under
 - *Complement*